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$$9 + 40i$$

$$= (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 &= 9 \\ 2xy &= 40 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 &= 9 \\ x^2 \cdot (-y^2) &= -400 \end{cases}$$

$$\lambda^2 - 9\lambda - 400 = 0$$

$$D = b^2 + 4ac = 81 + 1600 = 1681$$

$$\lambda_1 = \frac{-b - \sqrt{D}}{2a} = 25$$

$$\lambda_2 = \frac{-b + \sqrt{D}}{2a} = -16$$

$$x^2 = 25$$

$$-y^2 = -16$$

$$x = \pm 5 \text{ en } y = \pm 4$$

$$\text{Oplossing: } 5 + 4i \text{ of } -5 - 4i$$

References

- [1] template gebruikt van Oubayy Ahale met toestemming